

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

PRACTICAL VII

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Class: TYIT	Batch: 1
Date of Assignment: 01-08-2026	Date/Time of Submission: 01-08-2026

AIM :- Machine Learning

- 1) Using any small dataset (e.g., Iris or a CSV dataset), write a Python program to demonstrate the basic machine learning workflow.
 - a. Load dataset using Pandas.
 - b. Perform basic preprocessing (handling missing values or scaling).
 - c. Split the dataset into training and testing sets.
 - d. Train a simple classifier.
 - e. Display training and testing accuracy.

(iris dataset):-

Code:-

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, MinMaxScaler, StandardScaler
# Load Dataset
df = pd.read_csv("iris - iris - iris - iris.csv")
print("===== 1. LOAD THE DATASET =====")
print(df)
print("\nRitesh Chaubey")
# Display Dataset
print("\n===== 2. DISPLAY THE DATASET =====")
print(df)
print("\nRitesh Chaubey")
# Check Dataset Information
print("\n===== 3. CHECK DATASET INFORMATION =====")
df.info()
print("\nRitesh Chaubey")
# Display Dataset Shape
print("\n===== 4. DISPLAY DATASET SHAPE =====")
```

```

print(df.shape)
print("\nRitesh Chaubey")
# Display Column Names
print("\n===== 5. DISPLAY COLUMN NAMES =====")
print(df.columns)
print("\nRitesh Chaubey")
# Display Summary Statistics
print("\n===== 6. DISPLAY SUMMARY STATISTICS =====")
print(df.describe())
print("\nRitesh Chaubey")
# Check Data Types
print("\n===== 7. CHECK DATA TYPES =====")
print(df.dtypes)
print("\nRitesh Chaubey")
# Check Missing Values
print("\n===== 8. CHECK MISSING VALUES =====")
print(df.isnull().sum())
print("\nRitesh Chaubey")
# Handle Missing Values
print("\n===== 9. HANDLE MISSING VALUES =====")
numeric_columns = df.select_dtypes(include=["number"]).columns
df[numeric_columns] = df[numeric_columns].fillna(df[numeric_columns].mean())
print(df.isnull().sum())
print("\nRitesh Chaubey")
# Remove Duplicate Records
print("\n===== 10. REMOVE DUPLICATE RECORDS =====")
df.drop_duplicates(inplace=True)
print(df)
print("\nRitesh Chaubey")

# Encode Categorical Data
print("\n===== 11. ENCODE CATEGORICAL DATA =====")
encoder = LabelEncoder()
target_column = df.select_dtypes(include=["object"]).columns[-1]
df[target_column] = encoder.fit_transform(df[target_column])
print(df)
print("\nRitesh Chaubey")
# Rename Columns
print("\n===== 12. RENAME COLUMNS =====")
df.rename(columns={
    "sepal_length": "Sepal_Length",
    "sepal_width": "Sepal_Width",
    "petal_length": "Petal_Length",

```

```

    "petal_width": "Petal_Width",
    "species": "Species"
}, inplace=True)
print(df)
print("\nRitesh Chaubey")
# Drop Unnecessary Columns
print("\n===== 13. DROP UNNECESSARY COLUMNS =====")
df.drop(columns=[], inplace=True)
print(df)
print("\nRitesh Chaubey")
# Select Features
print("\n===== 14. SELECT FEATURES =====")
X = df.drop("Species", axis=1)
print(X)
print("\nRitesh Chaubey")
# Create New Feature
print("\n===== 15. CREATE NEW FEATURE =====")
df["Total_Length"] = df["Sepal_Length"] + df["Petal_Length"]
print(df)
print("\nRitesh Chaubey")
# Convert Data Types
print("\n===== 16. CONVERT DATA TYPES =====")

df["Species"] = df["Species"].astype(int)
print(df.dtypes)
print("\nRitesh Chaubey")
# Normalize Data (Min-Max Scaling)
print("\n===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====")
X = df.drop("Species", axis=1)
scaler = MinMaxScaler()
normalized_data = scaler.fit_transform(X)
print(normalized_data)
print("\nRitesh Chaubey")
# Standardize Data (Standard Scaling)
print("\n===== 18. STANDARDIZE DATA (STANDARD SCALING) =====")
standard_scaler = StandardScaler()
standardized_data = standard_scaler.fit_transform(X)
print(standardized_data)
print("\nRitesh Chaubey")
# Split Features and Target Variable
print("\n===== 19. SPLIT FEATURES AND TARGET VARIABLE =====")
X = df.drop("Species", axis=1)
y = df["Species"]

```

```

print("Features (X):")
print(X)
print("\nTarget (y):")
print(y)
print("\nRitesh Chaubey")

```

Output:-

```

===== 1. LOAD THE DATASET =====
   sepal_length  sepal_width  petal_length  petal_width  species
0             5.1           3.5           1.4           0.2    setosa
1             4.9           3.0           1.4           0.2    setosa
2             4.7           3.2           1.3           0.2    setosa
3             4.6           3.1           1.5           0.2    setosa
4             5.0           3.6           1.4           0.2    setosa
..           ...           ...           ...           ...     ...
145            6.7           3.0           5.2           2.3  virginica
146            6.3           2.5           5.0           1.9  virginica
147            6.5           3.0           5.2           2.0  virginica
148            6.2           3.4           5.4           2.3  virginica
149            5.9           3.0           5.1           1.8  virginica

```

```
[150 rows x 5 columns]
```

```
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```

```

   sepal_length  sepal_width  petal_length  petal_width  species
2             4.7           3.2           1.3           0.2    setosa
3             4.6           3.1           1.5           0.2    setosa
4             5.0           3.6           1.4           0.2    setosa
..           ...           ...           ...           ...     ...
145            6.7           3.0           5.2           2.3  virginica
146            6.3           2.5           5.0           1.9  virginica
147            6.5           3.0           5.2           2.0  virginica
148            6.2           3.4           5.4           2.3  virginica
149            5.9           3.0           5.1           1.8  virginica

```

```
[150 rows x 5 columns]
```

```
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```

```
===== 3. CHECK DATASET INFORMATION =====
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	sepal_length	150 non-null	float64
1	sepal_width	150 non-null	float64
2	petal_length	150 non-null	float64
3	petal_width	150 non-null	float64
4	species	150 non-null	str

```
dtypes: float64(4), str(1)
```

```
memory usage: 6.0 KB
```

```
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```

```
===== 4. DISPLAY DATASET SHAPE =====
```

```
(150, 5)
```

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```
===== 5. DISPLAY COLUMN NAMES =====
```

```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',  
      'species'],  
      dtype='str')
```

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```
===== 6. DISPLAY SUMMARY STATISTICS =====
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

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```
===== 7. CHECK DATA TYPES =====
```

```
sepal_length    float64  
sepal_width     float64  
petal_length    float64  
petal_width     float64  
species         str  
dtype: object
```

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```
===== 8. CHECK MISSING VALUES =====
```

```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

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```
===== 9. HANDLE MISSING VALUES =====
```

```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

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```
===== 10. REMOVE DUPLICATE RECORDS =====
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
..	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
[147 rows x 5 columns]
```

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```

===== 11. ENCODE CATEGORICAL DATA =====
      sepal_length  sepal_width  petal_length  petal_width  species
0           5.1           3.5           1.4           0.2           0
1           4.9           3.0           1.4           0.2           0
2           4.7           3.2           1.3           0.2           0
3           4.6           3.1           1.5           0.2           0
4           5.0           3.6           1.4           0.2           0
..           ...           ...           ...           ...           ...
145          6.7           3.0           5.2           2.3           2
146          6.3           2.5           5.0           1.9           2
147          6.5           3.0           5.2           2.0           2
148          6.2           3.4           5.4           2.3           2
149          5.9           3.0           5.1           1.8           2

[147 rows x 5 columns]

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```

```

===== 12. RENAME COLUMNS =====
      Sepal_Length  Sepal_Width  Petal_Length  Petal_Width  Species
0           5.1           3.5           1.4           0.2           0
1           4.9           3.0           1.4           0.2           0
2           4.7           3.2           1.3           0.2           0
3           4.6           3.1           1.5           0.2           0
4           5.0           3.6           1.4           0.2           0
..           ...           ...           ...           ...           ...
145          6.7           3.0           5.2           2.3           2
146          6.3           2.5           5.0           1.9           2
147          6.5           3.0           5.2           2.0           2
148          6.2           3.4           5.4           2.3           2
149          5.9           3.0           5.1           1.8           2

[147 rows x 5 columns]

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```

===== 13. DROP UNNECESSARY COLUMNS =====

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
..	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

[147 rows x 5 columns]

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===== 14. SELECT FEATURES =====

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
..	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

[147 rows x 4 columns]

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===== 15. CREATE NEW FEATURE =====

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species	\
0	5.1	3.5	1.4	0.2	0	
1	4.9	3.0	1.4	0.2	0	
2	4.7	3.2	1.3	0.2	0	
3	4.6	3.1	1.5	0.2	0	
4	5.0	3.6	1.4	0.2	0	
..	...	...	...	...	...	
145	6.7	3.0	5.2	2.3	2	
146	6.3	2.5	5.0	1.9	2	
147	6.5	3.0	5.2	2.0	2	
148	6.2	3.4	5.4	2.3	2	
149	5.9	3.0	5.1	1.8	2	

	Total_Length
0	6.5
1	6.3
2	6.0
3	6.1
4	6.4
..	...
145	11.9
146	11.3
147	11.7
148	11.6
149	11.0

===== 16. CONVERT DATA TYPES =====

```
Sepal_Length    float64
Sepal_Width     float64
Petal_Length    float64
Petal_Width     float64
Species         int64
Total_Length    float64
dtype: object
```

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===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====

```
[0.22222222 0.625      0.06779661 0.04166667 0.11956522]
[0.16666667 0.41666667 0.06779661 0.04166667 0.09782609]
[0.11111111 0.5        0.05084746 0.04166667 0.06521739]
[0.08333333 0.45833333 0.08474576 0.04166667 0.07608696]
[0.19444444 0.66666667 0.06779661 0.04166667 0.10869565]
[0.30555556 0.79166667 0.11864407 0.125      0.18478261]
[0.08333333 0.58333333 0.06779661 0.08333333 0.06521739]
[0.19444444 0.58333333 0.08474576 0.04166667 0.11956522]
[0.02777778 0.375      0.06779661 0.04166667 0.04347826]
[0.16666667 0.45833333 0.08474576 0.         0.10869565]
[0.30555556 0.70833333 0.08474576 0.04166667 0.16304348]
[0.13888889 0.58333333 0.10169492 0.04166667 0.10869565]
[0.13888889 0.41666667 0.06779661 0.         0.08695652]
[0.         0.41666667 0.01694915 0.         0.         ]
[0.41666667 0.83333333 0.03389831 0.04166667 0.17391304]
```

```

===== 18. STANDARDIZE DATA (STANDARD SCALING) =====
[[-0.91550949  1.0199714 -1.35773692 -1.33570001 -1.25165163]
 [-1.15755966 -0.12808217 -1.35773692 -1.33570001 -1.33145765]
 [-1.39960983  0.33113926 -1.41477817 -1.33570001 -1.45116669]
 [-1.52063491  0.10152855 -1.30069567 -1.33570001 -1.41126368]
 [-1.03653457  1.24958212 -1.35773692 -1.33570001 -1.29155464]
 [-0.55243423  1.93841426 -1.18661318 -1.07090176 -1.01223356]
 [-1.52063491  0.79036069 -1.35773692 -1.20330089 -1.45116669]
 [-1.03653457  0.79036069 -1.30069567 -1.33570001 -1.25165163]
 [-1.76268509 -0.35769288 -1.35773692 -1.33570001 -1.53097271]
 [-1.15755966  0.10152855 -1.30069567 -1.46809914 -1.29155464]
 [-0.55243423  1.47919283 -1.30069567 -1.33570001 -1.09203958]
 [-1.27858474  0.79036069 -1.24365442 -1.33570001 -1.29155464]
 [-1.27858474 -0.12808217 -1.35773692 -1.46809914 -1.37136067]
 [-1.88371017 -0.12808217 -1.52886066 -1.46809914 -1.69058476]
 [-0.06833389  2.16802498 -1.47181942 -1.33570001 -1.05213657]
 [-0.18935898  3.08646783 -1.30069567 -1.07090176 -0.97233054]
 [-0.55243423  1.93841426 -1.41477817 -1.07090176 -1.1718456 ]
 [-0.91550949  1.0199714 -1.35773692 -1.20330089 -1.25165163]
 [-0.18935898  1.70880355 -1.18661318 -1.20330089 -0.89252452]

```

```

===== 19. SPLIT FEATURES AND TARGET VARIABLE =====

```

Features (X):

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Total_Length
0	5.1	3.5	1.4	0.2	6.5
1	4.9	3.0	1.4	0.2	6.3
2	4.7	3.2	1.3	0.2	6.0
3	4.6	3.1	1.5	0.2	6.1
4	5.0	3.6	1.4	0.2	6.4
..	...	...	...	...	...
145	6.7	3.0	5.2	2.3	11.9
146	6.3	2.5	5.0	1.9	11.3
147	6.5	3.0	5.2	2.0	11.7
148	6.2	3.4	5.4	2.3	11.6
149	5.9	3.0	5.1	1.8	11.0

[147 rows x 5 columns]

[147 rows x 5 columns]

Target (y):

0	0
1	0
2	0
3	0
4	0
..	
145	2
146	2
147	2
148	2
149	2

Name: Species, Length: 147, dtype: int64

(supermarket)

Code:-

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, MinMaxScaler, StandardScaler
# Load Dataset
df = pd.read_csv("supermarket_sales - supermarket_sales.csv")
print("===== 1. LOAD THE DATASET =====")
print(df)
print("\nRitesh Chaubey")
# Display Dataset
print("\n===== 2. DISPLAY THE DATASET =====")
print(df)
print("\nRitesh Chaubey")
# Check Dataset Information
print("\n===== 3. CHECK DATASET INFORMATION =====")
df.info()
print("\nRitesh Chaubey")
# Display Dataset Shape
print("\n===== 4. DISPLAY DATASET SHAPE =====")
print(df.shape)
print("\nRitesh Chaubey")
# Display Column Names
print("\n===== 5. DISPLAY COLUMN NAMES =====")
print(df.columns)
print("\nRitesh Chaubey")
# Display Summary Statistics
print("\n===== 6. DISPLAY SUMMARY STATISTICS =====")
print(df.describe())
print("\nRitesh Chaubey")
# Check Data Types
print("\n===== 7. CHECK DATA TYPES =====")
print(df.dtypes)
print("\nRitesh Chaubey")
# Check Missing Values
print("\n===== 8. CHECK MISSING VALUES =====")
print(df.isnull().sum())
print("\nRitesh Chaubey")
# Handle Missing Values
print("\n===== 9. HANDLE MISSING VALUES =====")
numeric_columns = df.select_dtypes(include=["number"]).columns
df[numeric_columns] = df[numeric_columns].fillna(df[numeric_columns].mean())
print(df.isnull().sum())
print("\nRitesh Chaubey")
```

```

# Remove Duplicate Records
print("\n===== 10. REMOVE DUPLICATE RECORDS =====")
df.drop_duplicates(inplace=True)
print(df)
print("\nRitesh Chaubey")
# Encode Categorical Data
print("\n===== 11. ENCODE CATEGORICAL DATA =====")
encoder = LabelEncoder()
categorical_columns = df.select_dtypes(include=["object"]).columns
for col in categorical_columns:
    df[col] = encoder.fit_transform(df[col])
print(df)
print("\nRitesh Chaubey")
# Rename Columns
print("\n===== 12. RENAME COLUMNS =====")
df.rename(columns={
    "Invoice ID": "Invoice_ID",
    "Customer type": "Customer_Type",
    "Product line": "Product_Line",
    "Unit price": "Unit_Price",
    "Tax 5%": "Tax_5",
    "gross margin percentage": "Gross_Margin_Percentage",
    "gross income": "Gross_Income"
}, inplace=True)
print(df)
print("\nRitesh Chaubey")
# Drop Unnecessary Columns
print("\n===== 13. DROP UNNECESSARY COLUMNS =====")
df.drop(columns=["Invoice_ID"], inplace=True)
print(df)
print("\nRitesh Chaubey")
# Select Features
print("\n===== 14. SELECT FEATURES =====")
X = df.drop("Payment", axis=1)
print(X)
print("\nRitesh Chaubey")
# Create New Feature
print("\n===== 15. CREATE NEW FEATURE =====")
df["Sales_Per_Item"] = df["Total"] / df["Quantity"]
print(df)
print("\nRitesh Chaubey")
# Convert Data Types
print("\n===== 16. CONVERT DATA TYPES =====")

```

```

df["Quantity"] = df["Quantity"].astype(int)
df["Rating"] = df["Rating"].astype(float)
print(df.dtypes)
print("\nRitesh Chaubey")
# Normalize Data (Min-Max Scaling)
print("\n===== 17. NORMALIZE DATA (MIN-MAX SCALING) =====")
X = df.drop("Payment", axis=1)
scaler = MinMaxScaler()
normalized_data = scaler.fit_transform(X)
print(normalized_data)
print("\nRitesh Chaubey")
# Standardize Data (Standard Scaling)
print("\n===== 18. STANDARDIZE DATA (STANDARD SCALING) =====")
standard_scaler = StandardScaler()
standardized_data = standard_scaler.fit_transform(X)
print(standardized_data)
print("\nRitesh Chaubey")
# Split Features and Target Variable
print("\n===== 19. SPLIT FEATURES AND TARGET VARIABLE =====")
X = df.drop("Payment", axis=1)
y = df["Payment"]
print("Features (X):")
print(X)
print("\nTarget (y):")
print(y)
print("\nRitesh Chaubey")

```

Output:-

```

===== 1. LOAD THE DATASET =====

```

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	
..	...	...	...	...	...	
995	233-67-5758	C	Naypyitaw	Normal	Male	
996	303-96-2227	B	Mandalay	Normal	Female	
997	727-02-1313	A	Yangon	Member	Male	
998	347-56-2442	A	Yangon	Normal	Male	
999	849-09-3807	A	Yangon	Member	Female	

	Product line	Unit price	Quantity	Tax 5%	Total	\
0	Health and beauty	74.69	7	26.1415	548.9715	
1	Electronic accessories	15.28	5	3.8200	80.2200	
2	Home and lifestyle	46.33	7	16.2155	340.5255	
3	Health and beauty	58.22	8	23.2880	489.0480	
4	Sports and travel	86.31	7	30.2085	634.3785	
..	...	...	...	...	...	
995	Health and beauty	40.35	1	2.0175	42.3675	
996	Home and lifestyle	97.38	10	48.6900	1022.4900	
997	Food and beverages	31.84	1	1.5920	33.4320	
998	Home and lifestyle	65.82	1	3.2910	69.1110	
999	Fashion accessories	88.34	7	30.9190	649.2990	

===== 2. DISPLAY THE DATASET =====

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	
..	...	...	...	...	...	
995	233-67-5758	C	Naypyitaw	Normal	Male	
996	303-96-2227	B	Mandalay	Normal	Female	
997	727-02-1313	A	Yangon	Member	Male	
998	347-56-2442	A	Yangon	Normal	Male	
999	849-09-3807	A	Yangon	Member	Female	

	Product line	Unit price	Quantity	Tax 5%	Total	\
0	Health and beauty	74.69	7	26.1415	548.9715	
1	Electronic accessories	15.28	5	3.8200	80.2200	
2	Home and lifestyle	46.33	7	16.2155	340.5255	
3	Health and beauty	58.22	8	23.2880	489.0480	
4	Sports and travel	86.31	7	30.2085	634.3785	
..	...	...	...	...	...	
995	Health and beauty	40.35	1	2.0175	42.3675	
996	Home and lifestyle	97.38	10	48.6900	1022.4900	
997	Food and beverages	31.84	1	1.5920	33.4320	
998	Home and lifestyle	65.82	1	3.2910	69.1110	

===== 3. CHECK DATASET INFORMATION =====

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Invoice ID                            1000 non-null   str
1   Branch                               1000 non-null   str
2   City                                 1000 non-null   str
3   Customer type                        1000 non-null   str
4   Gender                               1000 non-null   str
5   Product line                         1000 non-null   str
6   Unit price                           1000 non-null   float64
7   Quantity                             1000 non-null   int64
8   Tax 5%                              1000 non-null   float64
9   Total                               1000 non-null   float64
10  Date                                 1000 non-null   str
11  Time                                 1000 non-null   str
12  Payment                             1000 non-null   str
13  cogs                                1000 non-null   float64
14  gross margin percentage              1000 non-null   float64
15  gross income                         1000 non-null   float64
16  Rating                              1000 non-null   float64
dtypes: float64(7), int64(1), str(9)
memory usage: 132.9 KB
```

```
===== 4. DISPLAY DATASET SHAPE =====  
(1000, 17)
```

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```
===== 5. DISPLAY COLUMN NAMES =====
```

```
Index(['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender',  
      'Product line', 'Unit price', 'Quantity', 'Tax 5%', 'Total', 'Date',  
      'Time', 'Payment', 'cogs', 'gross margin percentage', 'gross income',  
      'Rating'],  
      dtype='str')
```

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```
===== 6. DISPLAY SUMMARY STATISTICS =====
```

	Unit price	Quantity	Tax 5%	Total	cogs \
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	55.672130	5.510000	15.379369	322.966749	307.58738
std	26.494628	2.923431	11.708825	245.885335	234.17651
min	10.080000	1.000000	0.508500	10.678500	10.17000
25%	32.875000	3.000000	5.924875	124.422375	118.49750
50%	55.230000	5.000000	12.088000	253.848000	241.76000
75%	77.935000	8.000000	22.445250	471.350250	448.90500
max	99.960000	10.000000	49.650000	1042.650000	993.00000

	gross margin percentage	gross income	Rating
count	1000.000000	1000.000000	1000.000000
mean	4.761905	15.379369	6.97270
std	0.000000	11.708825	1.71958

```
===== 7. CHECK DATA TYPES =====
```

Invoice ID	str
Branch	str
City	str
Customer type	str
Gender	str
Product line	str
Unit price	float64
Quantity	int64
Tax 5%	float64
Total	float64
Date	str
Time	str
Payment	str
cogs	float64
gross margin percentage	float64
gross income	float64
Rating	float64
dtype:	object

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===== 8. CHECK MISSING VALUES =====

Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

dtype: int64

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===== 9. HANDLE MISSING VALUES =====

Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

dtype: int64

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===== 10. REMOVE DUPLICATE RECORDS =====

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	
..	...	...	...	...	...	
995	233-67-5758	C	Naypyitaw	Normal	Male	
996	303-96-2227	B	Mandalay	Normal	Female	
997	727-02-1313	A	Yangon	Member	Male	
998	347-56-2442	A	Yangon	Normal	Male	
999	849-09-3807	A	Yangon	Member	Female	

	Product line	Unit price	Quantity	Tax 5%	Total	\
0	Health and beauty	74.69	7	26.1415	548.9715	
1	Electronic accessories	15.28	5	3.8200	80.2200	
2	Home and lifestyle	46.33	7	16.2155	340.5255	
3	Health and beauty	58.22	8	23.2880	489.0480	
4	Sports and travel	86.31	7	30.2085	634.3785	

===== 11. ENCODE CATEGORICAL DATA =====

	Invoice ID	Branch	City	Customer type	Gender	Product line	\
0	814	0	2	0	0	3	
1	142	2	1	1	0	0	
2	653	0	2	1	1	4	
3	18	0	2	0	1	3	
4	339	0	2	1	1	5	
..	...	...	...	...	...	...	
995	153	2	1	1	1	3	
996	250	1	0	1	0	4	
997	767	0	2	0	1	2	
998	308	0	2	1	1	4	
999	935	0	2	0	0	1	

	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	\
0	74.69	7	26.1415	548.9715	26	146	2	522.83	
1	15.28	5	3.8200	80.2200	87	23	0	76.40	
2	46.33	7	16.2155	340.5255	81	155	1	324.31	
3	58.22	8	23.2880	489.0480	19	485	2	465.76	
4	86.31	7	30.2085	634.3785	57	29	2	604.17	

===== 12. RENAME COLUMNS =====

	Invoice_ID	Branch	City	Customer_Type	Gender	Product_Line	\
0	814	0	2	0	0	3	
1	142	2	1	1	0	0	
2	653	0	2	1	1	4	
3	18	0	2	0	1	3	
4	339	0	2	1	1	5	
..	...	...	...	...	...	...	
995	153	2	1	1	1	3	
996	250	1	0	1	0	4	
997	767	0	2	0	1	2	
998	308	0	2	1	1	4	
999	935	0	2	0	0	1	

	Unit_Price	Quantity	Tax_5	Total	Date	Time	Payment	cogs	\
0	74.69	7	26.1415	548.9715	26	146	2	522.83	
1	15.28	5	3.8200	80.2200	87	23	0	76.40	
2	46.33	7	16.2155	340.5255	81	155	1	324.31	
3	58.22	8	23.2880	489.0480	19	485	2	465.76	
4	86.31	7	30.2085	634.3785	57	29	2	604.17	

===== 13. DROP UNNECESSARY COLUMNS =====

	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	Quantity	\
0	0	2	0	0	3	74.69	7	
1	2	1	1	0	0	15.28	5	
2	0	2	1	1	4	46.33	7	
3	0	2	0	1	3	58.22	8	
4	0	2	1	1	5	86.31	7	
..	...	...	...	...	...	...	...	
995	2	1	1	1	3	40.35	1	
996	1	0	1	0	4	97.38	10	
997	0	2	0	1	2	31.84	1	
998	0	2	1	1	4	65.82	1	
999	0	2	0	0	1	88.34	7	

	Tax_5	Total	Date	Time	Payment	cogs	Gross_Margin_Percentage	\
0	26.1415	548.9715	26	146	2	522.83	4.761905	
1	3.8200	80.2200	87	23	0	76.40	4.761905	
2	16.2155	340.5255	81	155	1	324.31	4.761905	
3	23.2880	489.0480	19	485	2	465.76	4.761905	
4	30.2085	634.3785	57	29	2	604.17	4.761905	
..	...	...	...	...	...	...	...	
995	2.0175	42.3675	21	174	2	40.35	4.761905	
996	48.6900	1022.4900	70	342	2	973.80	4.761905	
997	1.5920	33.4320	58	154	0	31.84	4.761905	
998	3.2910	69.1110	45	261	0	65.82	4.761905	
999	30.9190	649.2990	40	160	0	618.38	4.761905	

===== 14. SELECT FEATURES =====

	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	Quantity	\
0	0	2	0	0	3	74.69	7	
1	2	1	1	0	0	15.28	5	
2	0	2	1	1	4	46.33	7	
3	0	2	0	1	3	58.22	8	
4	0	2	1	1	5	86.31	7	
..	...	...	...	...	...	...	...	...
995	2	1	1	1	3	40.35	1	
996	1	0	1	0	4	97.38	10	
997	0	2	0	1	2	31.84	1	
998	0	2	1	1	4	65.82	1	
999	0	2	0	0	1	88.34	7	

	Tax_5	Total	Date	Time	cogs	Gross_Margin_Percentage	\
0	26.1415	548.9715	26	146	522.83	4.761905	
1	3.8200	80.2200	87	23	76.40	4.761905	
2	16.2155	340.5255	81	155	324.31	4.761905	
3	23.2880	489.0480	19	485	465.76	4.761905	

===== 15. CREATE NEW FEATURE =====

	Branch	City	Customer_Type	Gender	Product_Line	Unit_Price	Quantity	\
0	0	2	0	0	3	74.69	7	
1	2	1	1	0	0	15.28	5	
2	0	2	1	1	4	46.33	7	
3	0	2	0	1	3	58.22	8	
4	0	2	1	1	5	86.31	7	
..	...	...	...	...	...	...	...	...
995	2	1	1	1	3	40.35	1	
996	1	0	1	0	4	97.38	10	
997	0	2	0	1	2	31.84	1	
998	0	2	1	1	4	65.82	1	
999	0	2	0	0	1	88.34	7	

	Tax_5	Total	Date	Time	Payment	cogs	Gross_Margin_Percentage	\
0	26.1415	548.9715	26	146	2	522.83	4.761905	
1	3.8200	80.2200	87	23	0	76.40	4.761905	
2	16.2155	340.5255	81	155	1	324.31	4.761905	
3	23.2880	489.0480	19	485	2	465.76	4.761905	
4	30.2085	634.3785	57	29	2	604.17	4.761905	

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[illegible]

Feature	Iris Dataset	Supermarket Dataset
Dataset Type	Flower dataset	Sales transaction dataset
Number of Rows	150	1000
Number of Columns	5	17
Target Variable	Species	Rating (for practice)
Data Type	Mostly Numeric	Numeric + Categorical
Missing Values	Usually None	Usually None
Duplicate Records	Rare	May Exist
Encoding Needed	Species only	Branch, City, Gender, Payment, etc.
Scaling Needed	Yes	Yes
Feature Engineering	Total_Length	Total_Amount
Unnecessary Columns	None	Invoice_ID
Machine Learning Task	Classification	Classification / Regression
Complexity	Simple	Moderate
Best Used For	Learning ML Basics	Data Preprocessing & Sales Analysis